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Intent-Based Networking with QoS-Aware Routing in SDN using AHP based AI Prioritization and Optimization

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***Abstract*—The increasing heterogeneity of network services in next-generation environments require intelligent and adaptive routing frameworks that can align with diverse and dynamic application level service requirements. Software-Defined Networking (SDN) decouples the control plane from data plane to shift the complexity from core network devices. Moreover, Intent-Based Networking (IBN), when integrated with SDN, provides a centralized and programmable platform to translate high level user intents into actionable network configurations. This paper proposes a hybrid quality of service (QoS)-aware routing framework that combines the Analytic Hierarchy Process (AHP) with adaptive Artificial Intelligence (AI) to support intent-driven decision-making in SDN. The AHP module interprets user-defined service intents into weighted QoS priorities, while the AI component dynamically refines these weights based on the network telemetry. This integration enables context-aware path selection that balances structure, adaptability, and explainability. Experimental results on real-world network topologies demonstrate that the proposed framework consistently outperforms benchmark and state-of-the-art approaches in terms of end-to-end (E2E) delay, packet loss, jitter, throughput, and intent satisfaction. Moreover, the system achieves fast re-routing convergence and maintains a low controller overhead, making it suitable for scalable, transparent, and high-performance deployment in IBN-compliant SDN infrastructures.**

***Index Terms*—Intent-Based Networking (IBN), Software-Defined Networking (SDN), Lightweight Artificial Intelligence (AI), Optimal Path Selection**

I. Introduction

THE explosive growth of heterogeneous network applications ranging from real-time video streaming, industrial automation, and cloud gaming to mission-critical services like remote healthcare in Internet of Medical Things (IoMT) and connected vehicles has placed unprecedented demands on modern communication infrastructures. These applications not only differ in bandwidth consumption but also in latency sensitivity, jitter tolerance, and reliability requirements [1], [2], [3]. As networks scale to accommodate billions of Internet of Things (IoT) devices and dynamic service contexts, the ability to intelligently and automatically adapt to varying Quality of Service (QoS) expectations has become essential. Moreover, traditional network architectures, built on vendor-specific hardware and static rule-based configurations, offer limited programmability and are inherently rigid. Their tightly coupled control and data planes lead to inefficient resource utilization, slow fault recovery, and operational complexity. In response, *Software-Defined Networking (SDN)* has emerged as a transformative solution [4], [5]. By decoupling the control logic from the underlying forwarding hardware in the data plane, SDN provides centralized visibility and control, enabling programmability, dynamic policy enforcement, and network-wide orchestration. However, even SDN centralized control is not inherently capable of understanding or interpreting the application-specific or user-defined service goals that arise in complex environments [6], [7].

Moreover, integrating Artificial Intelligence (AI) algorithms into Intent-Based Networking (IBN) systems presents a promising approach to autonomously fulfill user intents. These advanced algorithms can analyze the real-time state of the network and make intelligent decisions to optimize both network performance and service quality. For example, AI algorithms has demonstrated strong capabilities in optimizing resource allocation across various network domains; such as managing power and radio resources in the access domain, and allocating virtual computing and storage resources at the network edge. These optimized policies can then be enforced by the IBN system to ensure autonomous network operations that align with specified intent. As a result, these technologies can help IoT systems deliver innovative services while overcoming critical orchestration challenges [8], [9], [10], [11], [12], [13], [14].

However, there is still a need for advancements in IoT systems which needs intelligent as well as adaptive routing frameworks meeting dynamic service requirements with a support of intent driven decisions leveraging SDN, AI and decision making methods. To address this, IBN has gained significant traction as an abstraction layer that allows users and applications to specify high level service objectives such as ensure low latency for video traffic or prioritize industrial telemetry with minimal loss rather than low level configurations [15], [16]. These intents are translated by the network into concrete policies and flow rules from the control plane of SDN to the data plane. Despite its conceptual appeal, IBN introduces new challenges. Most of the existing IBN frameworks are limited in their ability to dynamically reconcile conflicting intents, prioritize among multiple QoS dimensions, or adapt to rapid changes in network state. Many rely on static policy mapping

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or manually tuned decision rules, which cannot effectively respond to real-time congestion, dynamic flow behavior, or topology updates. Furthermore, to address these limitations, we propose a novel, hybrid framework that integrates the *Analytic Hierarchy Process (AHP)* with lightweight *AI*-based optimization to enable intent-aware, QoS-driven routing within SDN environments. AHP is a well-established multi-criteria decision-making (MCDM) technique that enables systematic ranking of alternatives based on pairwise comparisons across multiple criteria [17], [18], [19]. In our framework, AHP is used to translate user intents into a weighted hierarchy of QoS attributes reflecting their relative importance for a specific service flow. These initial weights, however, may become suboptimal as network conditions evolve. Therefore, we augment the static AHP model with adaptive AI components capable of analyzing real-time telemetry and adjusting these weights dynamically. This fusion ensures that routing decisions are both transparent (via AHP) and responsive (via AI), making the system capable of maintaining intent fidelity while optimizing performance under diverse and dynamic conditions.

Our approach leverages real-time monitoring, feedback loops, and telemetry analytics to inform decision updates [20], [21]. The framework enables scalable and explainable service differentiation by autonomously adjusting routing policies in accordance with both user-defined intent structures and operational network realities. This is particularly advantageous in multi-tenant environments, edge-cloud deployments, and infrastructure supporting 5G/6G services, where traffic demands are highly dynamic and performance expectations vary significantly across applications.

The main contributions of this work are as follows:

- We propose a novel hybrid decision-making framework that integrates the interpretability of AHP with the adaptability of AI for real-time, intent-driven routing in SDN.
- We design a modular system architecture that includes intent parsing, multi-criteria QoS prioritization, AI-based weight adjustment, and flow rule deployment.
- We formulate intent scoring and dynamic path selection that accommodates evolving user intents and network states.
- We evaluate our framework against benchmark models including static AHP, fuzzy-AHP, IoT E2E, and reinforcement (RL)-based routing using real-world topologies (Abilene, USNet, OS3E, and DFN).
- We demonstrate that our approach significantly improves delay, jitter, packet loss, throughput, and intent satisfaction ratio, while ensuring rapid convergence and low computational overhead at the controller.

The remainder of the paper is organized as follows: Section II reviews related work in IBN, QoS-aware SDN routing, and AI-based optimization. Section III presents the system architecture, modeling and details of the routing algorithm. Section IV describes the experimental setup and performance metrics. Section V provides a comparative analysis and discussion. Finally, Section VI concludes the paper.

II. Related Work

This detailed literature review describes the progression and integration of IBN, QoS-aware routing, and AI techniques within SDN environments. The convergence of these domains highlights the necessity for hybrid frameworks that can dynamically adapt to evolving network demands while maintaining structured decision-making processes.

A. Advancements in IBN within SDN

IBN has gained increasing attention as a paradigm that enables the automation of network management by abstracting operational tasks into high level policy definitions. Rather than specifying configurations at the device or protocol level, IBN allows network operators to define desired outcomes, which are then automatically translated and enforced by the network infrastructure. Within the SDN context, this paradigm is especially powerful, as SDN offers centralized control, global network visibility, and programmable interfaces with ideal conditions for implementing intent translation and enforcement mechanisms. Jacobs et al. [22] proposed an innovative machine learning-based framework for intent refinement in self-driving networks. Their approach integrates operator feedback into a learning loop, enhancing the systems ability to interpret and execute user-defined intents with higher accuracy over time. While their work laid foundational steps toward autonomous intent management, the intent translation process still requires more robust mechanisms to accommodate diverse network scenarios and service types. Furthermore, many existing IBN frameworks focus on static or predefined policies, lacking adaptability when intent priorities or network conditions change dynamically. As a result, recent research has pivoted toward enhancing intent granularity, integrating QoS-awareness into the intent layer, and automating conflict resolution among overlapping intents. These efforts highlight the need for more flexible and intelligent IBN systems that can operate effectively in heterogeneous and rapidly evolving network environments.

B. QoS-Aware Routing Mechanisms in SDN

QoS assurance is critical in SDN environments that serve a wide range of applications with varying service-level requirements. Traditional routing protocols typically optimize for a single metric, such as hop count or static link cost, but this is insufficient for supporting multimedia, real-time, or mission-critical traffic. The programmability of SDN enables the design of customized routing mechanisms that can evaluate and prioritize multiple QoS metrics simultaneously, such as bandwidth availability, latency, jitter, and packet loss. Win and Mya [23] proposed a traffic classification-based method for QoS-aware path selection. Their strategy involves categorizing flows by traffic type and selecting optimal paths based on estimated link performance and congestion. While their approach achieves improvements in flow-level differentiation, it lacks dynamic re-evaluation under changing network loads. Kumar et al. [24] developed *fybrrLink*, a QoS-aware routing protocol tailored for SDN-enabled satellite-terrestrial networks. Their approach

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TABLE I
COMPARISON OF RELATED WORK WITH THE PROPOSED AHP-AI QOS-AWARE IBN FRAMEWORK

Approach	**Intent Translation Mechanism**	**QoS-Aware Routing Capability**	**AI Integration Strategy**	**Real-Time Adaptability**	**Multi-Criteria Decision Support**
Jacobs et al. [22]	ML-based refinement loop	No	Supervised Learning (Feedback-driven)	Partial (Post-training)	No
Win & Mya [23]	Not applicable	Traffic Class-Based Static Pathing	None	Limited (Rule-based)	No
Kumar et al. [24]	Not applicable	Yes (Satellite-aware)	Heuristic Learning (Static)	Partial (Scenario-dependent)	No
Rezaei & Ghaffari [25]	Not applicable	Yes	None	No	Yes (Fuzzy AHP)
Song et al. [26]	Not applicable	Yes	Deep Reinforcement Learning (DQN)	Yes (Model-trained)	No
Alenazi et al. [27]	Not applicable	Yes (IoT-aware)	Deep RL (Flow-Aware)	Yes	No
Wumian et al. [28]	Not applicable	Yes	Transferable Reinforcement Learning	Yes (Reusable Models)	No
Sun et al. [29]	Not applicable	Yes (Cross-domain MARL)	Multi-Agent Reinforcement Learning	Yes (Cooperative Agents)	No
Song et al. [26]	Not applicable	Yes	Deep Reinforcement Learning	Yes	No
Kang et al. [30]	Not applicable	Yes (Dragonfly Network)	Multi-Agent Reinforcement Learning	Yes (Distributed Agents)	No
Chen et al. [31]	Not applicable	Yes (Network Slicing)	MILP Optimization	Yes	Yes (QoS Constraints)
Shahriar et al. [32]	Not applicable	Yes (Edge Networks)	SDN-OpenFlow Based	Yes (Dynamic Provisioning)	Yes (Traffic Prioritization)
Alhilali & Montazerolghaem [33]	Not applicable	Yes	AI-Based Load Balancing	Yes	No
Ali et al. [34]	Not applicable	Yes (IoT E2E)	OAWM + SDN Hierarchy	Yes (Domain-aware)	Yes (Additive Weights + TSC Mapping)
AHP-AI Framework (Proposed)	AHP-based Intent Manager	**Yes (Multi-QoS Intent-Aware)**	AHP + Adaptive AI Optimizer	**Yes (Dynamic Reprioritization)**	**Yes (AHP + AI Integration)**

considers satellite link characteristics and SDN controller interactions to dynamically select the best communication paths, thereby improving throughput and reducing delay in highly variable environments.

More structurally, Rezaei and Ghaffari [25] introduced a fuzzy AHP-based routing framework in SDN. Their model combines expert judgment and fuzzy logic to weigh multiple QoS parameters, resulting in a multi-criteria routing decision process that better reflects real-world trade-offs. Their simulation results demonstrated notable gains in end-to-end (E2E) delay reduction and throughput stability. Despite these benefits, the reliance on static decision matrices and lack of real-time adaptability remain key challenges, especially in scenarios with frequent traffic shifts or topology changes. These findings underline the value of integrating decision support mechanisms with adaptive optimization in SDN-based QoS routing systems.

C. Integration of AI Techniques for QoS Optimization in SDN

The application of AI in SDN has emerged as a transformative approach to automate complex decision-making tasks, particularly for QoS-aware routing and traffic engineering. Unlike rule-based systems, AI techniques are capable of learning from experience and adapting to dynamic environments, making them highly suitable for modern, elastic network infrastructures. Song et al. [26] applied DRL to learn optimal routing paths that minimize delay while adapting to real-time congestion metrics. Their results show improved QoS under variable traffic conditions; however, their solution introduces high computational overhead due to the complexity of deep learning models. Alenazi et al. [27] extended this idea in the context of SD-IoT networks for precision agriculture, using DRL for flow-aware routing that accounts for energy efficiency and latency, demonstrating the models effectiveness in constrained environments. Similarly, Wumian et al. [28] proposed a reusable reinforcement learning model that can be transferred across different network scenarios, reducing training time and increasing adaptability. Furthermore, Ali

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et al. [34] propose a two-level SDN framework tailored for E2E QoS provisioning in heterogeneous IoT environments, specifically in precision agriculture. Their approach leverages an Optimal Additive Weighting Module (OAWM) within each SDN domain to select the best Traffic Service Class (TSC) based on multi-criteria QoS requirements (delay, jitter, PLR, bandwidth, and cost). At the global control level, a comprehensive mapping mechanism ensures the cumulative QoS along the E2E route meets sensor application needs. Their results show significant performance improvements in comparison with benchmark schemes, especially in delay-sensitive environments. This aligns with the current study's goals in leveraging intelligent routing strategies for intent-based, real-time QoS optimization.

D. Multi-Agent RL for Cross-Domain Routing

As SDN networks scale and span multiple administrative domains, routing decisions become increasingly complex due to the need for coordination across heterogeneous controllers, policies, and performance objectives [35]. Multi-agent reinforcement learning (MARL) has been proposed as a solution to this problem, allowing each domain or sub-controller to operate semi-independently while learning from shared experience and collective goals. Sun et al. [29] explored the application of MARL in cross-domain SDN routing. Their study compares competitive versus cooperative learning models, ultimately demonstrating that cooperative agentswhere controllers share partial state informationachieve better global QoS optimization without sacrificing local autonomy. The authors highlight that MARL can effectively balance scalability, policy coordination, and adaptive learning, making it a strong candidate for managing cross-domain service-level agreements (SLAs). However, the training complexity and communication overhead between agents must be carefully managed to prevent performance degradation.

E. Hybrid Decision-Making Models Combining AHP and AI

While the AHP method is widely regarded for its interpretability and rigor in handling multi-criteria decision problems, its static nature makes it less suitable for dynamic network environments. AHP typically requires manually constructed pairwise comparison matrices and does not adjust weights in real-time. To overcome these limitations, recent work has explored hybrid models that combine AHP with adaptive AI techniques to support both structured decision logic and real-time responsiveness. Rezaei and Ghaffari [25] presented one such hybrid approach by embedding fuzzy logic into the AHP decision process for SDN routing. While this enhanced the flexibility of weight assignments, it did not support continuous learning from evolving traffic conditions. More recent proposals suggest combining AHP with reinforcement learning or metaheuristic algorithms, allowing systems to start with a human-defined intent structure and adapt it as network conditions change. The hybrid approach not only improves decision quality but also enhances explainability by retaining a clear logic trail from user intents to final routing decisions. It provides a promising foundation for intelligent, policy-driven routing frameworks that can reconcile user expectations with dynamic operational realities.

III. Proposed Method

This section introduces the mathematical formulation and architectural design of the proposed hybrid framework for intent-based, QoS-aware routing in SDN. The system integrates analytic decision-making via the AHP with real-time adaptive optimization using lightweight AI methods. The primary goal is to ensure that routing decisions remain aligned with diverse, high level user intents under fluctuating network conditions, while preserving system transparency and scalability. Furthermore, the architecture, depicted in Figure 1, is composed of three core modules: the Intent Manager, the AHP-AI Engine, and the SDN controller. The Intent Manager receives abstract service-level goals (e.g., minimize latency for video traffic, maximize throughput for data analytics) from network operators or applications and converts them into measurable QoS objectives. These objectives typically defined over delay, packet loss ratio (PLR), jitter, and throughput are used by the AHP-AI engine to determine optimal paths through the network. The controller executes these decisions by deploying corresponding flow rules using OpenFlow or similar southbound APIs.

The high level goals of a network operator such as minimizing latency or maximizing throughput are articulated in natural language or policy-driven intents leveraging IBN. The IBN system leverages AI translation and policy abstraction approaches to map these intents into low-level network policies. These are then decomposed into specific rules, such as flow prioritization, QoS configurations, or traffic engineering constraints, which are sent to the controller via northbound API. The controller, with its global network view, converts these abstract rules into precise configurations for switches and routers using southbound API. Consequently, The IBN ensures the operators goals (e.g., latency minimization) are realized dynamically and consistently across the data plane.

At the core of this system, the AHP-AI engine executes a two-phase logic. In the first phase, the engine applies the AHP to prioritize multiple QoS parameters. The network topology is modeled as a directed graph $G(V, E)$ where V denotes SDN switches and E the set of links. For a given flow request from source s to destination d, let $P = \{p_1, p_2, ..., p_m\}$ be the set of feasible paths. Each path is evaluated with respect to n QoS criteria $C = \{c_1, c_2, ..., c_n\}$ using a pairwise comparison matrix $A = [a_{ij}]$, where $[a_{ij}]$ shows the values in the matrix:

$$a_{ij} = \frac{w_i}{w_j} \tag{1}$$

Here, w_i and w_j represent the relative importance of criteria c_i and c_j as assigned by the operator or application policy. The weights are normalized through:

$$w_i = \frac{1}{n} \sum_{j=1}^{n} \left(\frac{a_{ij}}{\sum_{k=1}^{n} a_{kj}} \right) \tag{2}$$

To validate consistency, we compute the Consistency Index (CI) and Consistency Ratio (CR) using:

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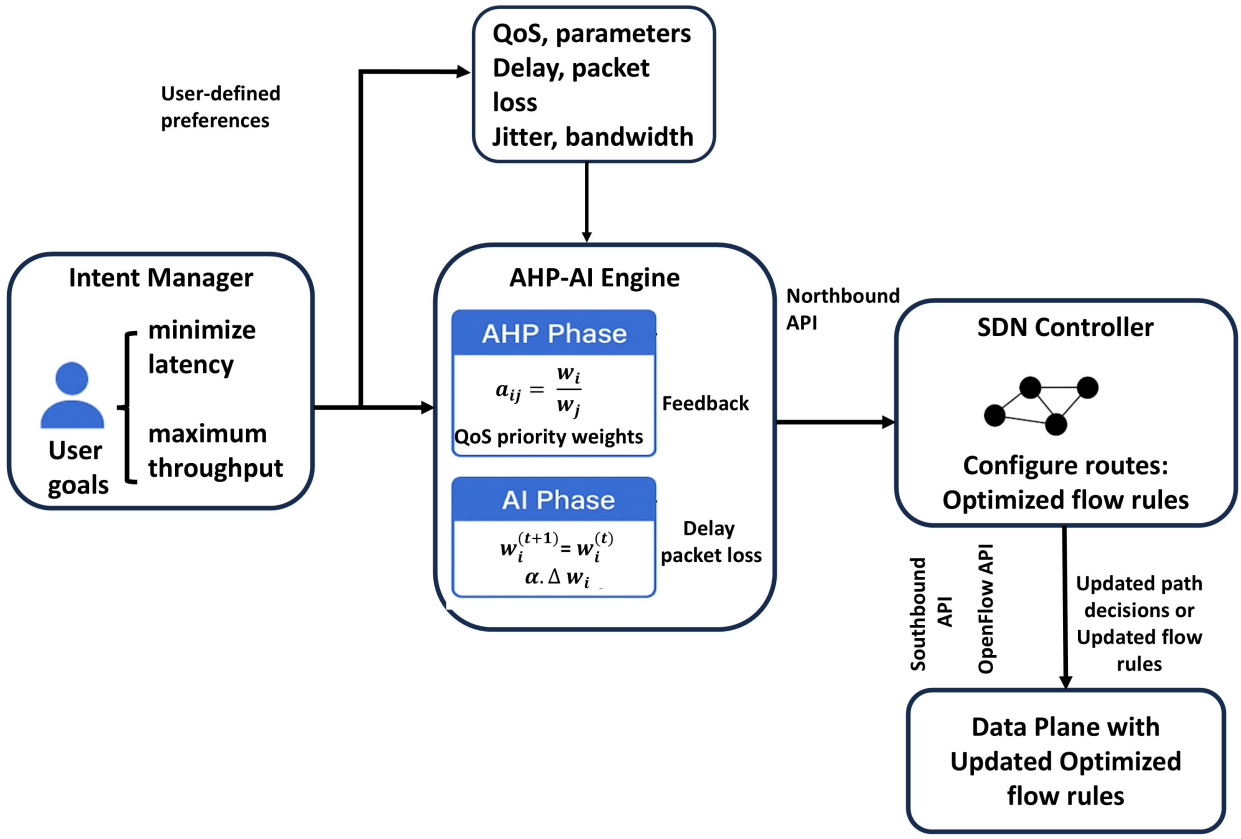

Fig. 1. Architecture of the proposed Intent-Based AHP-AI-based QoS-aware routing framework within a Software-Defined Networking environment.

$$CI = \frac{\lambda_{\max} - n}{n-1}, \quad CR = \frac{CI}{RI} \tag{3}$$

where $\lambda_{\max}$ is the maximum eigenvalue of A, and RI is the Ratio Index. If $CR \leq 0.1$, the weight structure is accepted i.e. it illustrates that the values are correct and consistent with respect to the comparison matrix; otherwise, it is discarded and re-evaluated to satisfy the condition. Furthermore, each path p_k is then scored via a weighted sum:

$$S(p_k) = \sum_{i=1}^{n} w_i \cdot q_i(p_k) \tag{4}$$

where $q_i(p_k)$ is the normalized performance score of path p_k with respect to criterion c_i, calculated as:

$$q_i(p_k) = \frac{c_i^{\max} - c_i(p_k)}{c_i^{\max} - c_i^{\min}} \tag{5}$$

This ensures comparability between heterogeneous metrics. The optimal path is selected as:

$$p^* = \arg\max_{p_k \in P} S(p_k) \tag{6}$$

To enable adaptability under dynamic network conditions, the second phase of the AHP-AI Engine implements AI-driven weight refinement. In its simplest form, weights are updated using a gradient-like rule:

$$w_i^{(t+1)} = w_i^{(t)} + \alpha \cdot \Delta w_i \tag{7}$$

Here, Δw_i captures the temporal deviation in criterion c_i due to network feedback, and α is a learning rate parameter. To avoid oscillation, weights are bounded and renormalized at each step. Further refinement is achieved using RL where the state s_t captures current network conditions, the action a_t represents a weight reallocation vector, and the reward $R(s_t, a_t)$ reflects performance gains. The learning objective is to maximize expected cumulative return:

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right] \tag{8}$$

where π_θ is the policy over weight updates parameterized by θ, and γ is a discount factor. The reward function is crafted to encourage improvements in intent satisfaction ratio (ISR) while penalizing degradations in key QoS parameters:

$$R(s_t, a_t) = \beta_1 \cdot \Delta\text{ISR}_t - \beta_2 \cdot \Delta\text{PLR}_t - \beta_3 \cdot \mathcal{L}_{\text{entropy}} \tag{9}$$

Here, ΔISR_t and ΔPLR_t reflect the temporal changes in ISR and PLR, and $\mathcal{L}_{\text{entropy}} = -\sum_i \pi_\theta(a_i|s_t) \log \pi_\theta(a_i|s_t)$ is an entropy regularization term used to balance exploration and stability. Finally, to discourage instability in network behavior due to frequent route flapping, a path-switching penalty $\psi(p_k)$ can be included in the score:

$$\tilde{S}(p_k) = S(p_k) - \lambda \cdot \psi(p_k) \tag{10}$$

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where $\psi(p_k)$ represents the switching frequency or historical load on p_k, and λ is a penalty factor. This extended scoring ensures long-term stability and fairness, especially in multi-tenant or delay-sensitive environments. Through combining the deterministic clarity of AHP with the adaptive intelligence of reinforcement-based learning, the proposed hybrid model facilitates intent-compliant and QoS-optimized routing that can adapt in real time to changes in network topology, congestion levels, and service objectives.

Furthermore, the proposed algorithm 1 for QoS-aware intent-driven routing in SDN comprises several sequential stages. Initially, the system performs *Intent Extraction*, wherein it captures high level service intents from users and translates them into structured QoS requirements. These intents typically express preferences such as minimizing latency, maximizing throughput, or balancing load across the network. Following this, the algorithm proceeds to *Criteria Definition and Normalization*. In this stage, a set of n QoS criteria $C = \{c_1, c_2, \ldots, c_n\}$ is defined. Each criterion represents a network performance metric such as bandwidth, delay, jitter, or packet loss. The values associated with these criteria are normalized across all candidate paths using standard techniques like min-max scaling or z-score normalization to ensure comparability. Next, the system applies the AHP for weight calculation. A pairwise comparison matrix A is constructed based on user preferences over the defined criteria. From this matrix, a priority vector $\vec{w}$ is computed as outlined in Eq. (2). To ensure the reliability of user preferences, the consistency of the matrix is evaluated using the Consistency Ratio (CR), calculated as shown in Eq. (3). With the normalized criteria and their respective weights, the algorithm performs *Path Scoring*. For each candidate path p_k, a composite score $S(p_k)$ is computed by aggregating the weighted QoS attributes according to Eq. (4). This score reflects the suitability of each path with respect to user-defined QoS priorities.

To further enhance adaptability, the algorithm incorporates an *AI-Based Weight Refinement* mechanism. Real-time telemetry datasuch as congestion levels, latency, and link reliabilityis collected from the network. An AI model, such as reinforcement learning or a genetic algorithm, is employed to refine the weight vector $\vec{w}$ dynamically. This process is mathematically represented by Eq. (7), which adjusts the weights in response to fluctuating network conditions. After updating the weights, the system proceeds to *Optimal Path Selection*, wherein the path p^* with the highest composite score is selected using Eq. (6). This ensures that the chosen path best satisfies the user's QoS intents under current network conditions. Finally, in the *Rule Installation* stage, the selected path p^* is instantiated in the network. The SDN controller programs the forwarding rules into the appropriate switches, ensuring that traffic is routed according to the optimized, intent-aware QoS path.

The proposed AHP-AI hybrid framework leverages structured human decision modeling with adaptive machine learning to achieve intent satisfaction in SDN routing. The combination of AHP-based initial prioritization with AI-based dynamic weight refinement provides both interpretability and responsiveness, offering a significant advancement over traditional static or AI-only routing systems.

Algorithm 1 Intent-Aware AHP-AI QoS Routing Algorithm

Require: Network Graph $G(V, E)$, User Intent I, QoS Criteria C, Set of Paths P
Ensure: Optimal Path p^*

```
Extract QoS parameters from intent I
Construct pairwise comparison matrix A
Compute priority weights w⃗ ← AHP(A)
Check consistency ratio CR
if CR > 0.1 then
    Request user to revise preferences
    For revising preferences, go step 2
end if
for each path p_k ∈ P do
    Normalize QoS metrics q_i(p_k) for all i
    Compute score S(p_k) ← Σ w_i · q_i(p_k)
end for
Collect network feedback F (delay, congestion, etc.)
Update weights: w_i ← w_i + α · Δw_i(F)
Re-score all paths using updated weights
Select p* = arg max S(p_k)
Updated flow rules on SDN controller
Send the flow rules via southbound API
Install the flow rules in the data plane
return p*
```

IV. Experimental Setup and Evaluation Metrics

This section details the simulation setup and performance metrics used to evaluate the proposed AHP-AI-based QoS-aware routing framework in SDN. The experiments are designed to benchmark our approach against recent state-of-the-art IBN and SDN routing methods from the literature, while demonstrating its superior adaptability, intent satisfaction, and QoS performance.

A. Simulation Environment

To evaluate the effectiveness of the proposed AHP-AI-based framework for intent-driven, QoS-aware routing in SDN, a comprehensive simulation environment was constructed. The emulation was carried out using Mininet, consistent with the methodologies used in prior studies such as Oliveira et al. [36]. For network topologies, we adopted realistic backbone structures sourced from the Internet Topology Zoo, particularly the Abilene, DFN and OS3E networks, which have been commonly used to benchmark routing algorithms in SDN [24].

Traffic was generated using widely accepted tools such as IPerf and distributed Internet traffic generator (D-ITG) as illustrated in [35], enabling the evaluation of bandwidth consumption, delay variation, and jitter under controlled congestion levels. This mirrors the traffic modeling practices seen in recent QoS optimization studies, including DRL-based adaptive routing for IoT scenarios by Alenazi et al. [27] and dynamic edge network simulations in Shahriar et al. [32]. In addition to flow emulation, custom northbound APIs were developed to link the AHP-AI engine with the SDN

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controllers decision modules. In addition, the flow rule updation and modification was implemented via OpenFlow 1.3, ensuring compatibility with most research-grade controllers. The architecture was designed to support real-time telemetry collection and feedback-driven adjustments, similar to the reinforcement learning environments discussed in the multi-agent routing schemes proposed by Kang et al. [30] and Sun et al. [29]. This setup allowed the proposed system to simulate dynamic prioritization and policy enforcement mechanisms under conditions approximating real operational scenarios.

B. Comparison Baselines

To rigorously assess the performance of our proposed hybrid decision-making framework, we conducted comparative analysis against a range of baseline models that are well-recognized in the literature. As a fundamental benchmark, we implemented routing based on Dijkstra's algorithm, which, while efficient, does not account for QoS metrics and thus serves as a useful lower-bound reference [37]. We also included a static AHP-based routing model derived from earlier SDN studies like those by Ali and Roh [38], where user preferences are encoded into decision matrices but not updated over time. This approach highlights the limitations of fixed decision logic in dynamic environments. To assess the benefits of learning-based adaptation, we incorporated DRL-based routing using models such as Deep Q-Networks (DQN), similar to those proposed by Song et al. [26] and further enhanced in IoT-aware scenarios by Alenazi et al. [27]. Additionally, Wumian et al. [28] contributed a transferable reinforcement learning strategy capable of reusing learned policies across network conditions, which we used to evaluate cross-scenario adaptability.

For static multi-criteria decision-making, we evaluated the fuzzy AHP-based model proposed by Rezaei and Ghaffari [25], which introduces fuzziness into weight generation but still operates without feedback. We further extended our baseline set by including recent models such as the MILP-based QoS routing mechanism in network slicing by Chen et al. [31], and the prioritized multi-tenant traffic engineering framework from Shahriar et al. [32], both of which incorporate optimization logic relevant to multi-domain service provisioning. In addition, we incorporated the two level SDN architecture by Ali et al. [34], which applies an OAWM for selecting traffic service classes in heterogeneous IoT environments. Their model demonstrates effective E2E QoS control through domain level and global coordination, offering a compelling benchmark for service-aware and adaptive routing under constrained sensor network conditions. Together, these baselines provide a comprehensive benchmark for evaluating our framework in terms of delay minimization, packet loss, throughput, responsiveness, and multi-criteria optimization capability, under conditions reflecting real-world diversity and volatility in traffic behavior.

C. Evaluation Metrics

We evaluate our framework using key metrics derived from prior work, with an emphasis on both traditional QoS indicators and novel intent-aware metrics:

- **E2E Delay (ms):** Measures the average latency across all active flows. A key parameter for real-time applications, also considered in [26].
- **Packet Loss Ratio (PLR):** Defined as the ratio of dropped packets to total packets sent. Critical for assessing reliability [23].
- **Jitter (ms):** Assesses variation in packet arrival intervals, relevant for streaming and VoIP scenarios [28].
- **Throughput (Mbps):** Represents aggregate data transfer rates between communicating endpoints.
- **Intent Satisfaction Ratio (ISR):** A novel metric defined as:
$$ISR = \frac{\text{Number of Intents Satisfied}}{\text{Total Intents Submitted}} \tag{11}$$
This metric captures the success rate of fulfilling user-defined QoS goals highlighting how well a framework respects user intent, which is not explicitly evaluated in most traditional SDN works but emphasized in IBN studies like [22], [29].
- **Re-routing Convergence Time (ms):** The average time taken to update paths in response to QoS degradation or link failures. It reflects system responsiveness and is key for adaptive frameworks.
- **Controller Overhead:** CPU and memory usage on the controller, indicating scalability and resource efficiency. Benchmarked similar to [27].

This experimental design ensures that our evaluation is comprehensive and grounded in current academic benchmarks, while the ISR metric uniquely highlights our models alignment with user-defined intentssomething most existing models do not explicitly address.

V. Results and Discussion

This section presents the evaluation of the proposed AHP-AI-based intent-aware routing framework, benchmarked against several established and recent routing models. The comparative analysis includes traditional baselines including: IoT E2E, Static AHP, fuzzy-AHP, and various AI-enhanced routing approaches such as DRL, reusable RL, MARL for cross-domain routing, and MILP-based QoS-aware network slicing. The goal is to demonstrate not only the performance superiority of the proposed system but also its adaptability, scalability, and interpretability under diverse service demands and network conditions.

Table II shows results for Abilene topology. i.e. with delay 20.8 ms, controller overhead 11.3, convergence as 58.4 ms, ISR ratio is 93.4, a throughput of 95 Mbps, jitter is 2.4, and PLR is 0.63. Similarly, Table III summarizes the performance metrics across seven critical dimensions for the baseline US-Net topology. The proposed method delivers the best or near-best performance in all categories, particularly excelling in delay (23.4 ms), packet loss ratio (0.72%), and ISR at 92.3%. This is a direct result of combining the structured prioritization of AHP with the real-time adaptive intelligence of AI, enabling both responsive and explainable QoS routing.

To further evaluate robustness, we analyze the framework across four real-world network topologies of increasing complexity: OS3E, and DFN. The performance metrics across

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TABLE II
ROUTING MODEL PERFORMANCE ON ABILENE TOPOLOGY (11 NODES, 14 EDGES)

Routing Model	Delay (ms)	PLR (%)	Jitter (ms)	Throughput (Mbps)	ISR (%)	Convergence (ms)	Controller Overhead
IoT E2E [34]	25.3	0.93	2.7	93.3	88.1	67.2	12.1
Static AHP [38]	28.1	1.74	3.9	92.7	80.2	96.1	10.5
Fuzzy AHP [25]	26.4	1.22	3.7	93.1	84.3	88.5	12.2
Deep RL (DQN) [26]	23.9	1.01	3.1	95.1	86.0	73.1	17.1
Flow-aware DRL [27]	22.6	0.86	2.9	94.5	87.5	66.4	18.6
Reusable RL [28]	23.5	0.94	3.0	93.7	85.6	71.2	15.5
MARL (Sun et al.) [29]	22.2	0.89	2.7	93.9	87.8	65.3	16.1
MARL (Kang et al.) [30]	21.7	0.83	2.5	94.3	89.4	62.4	15.4
MILP QoS-Slicing [31]	23.5	1.06	3.3	93.0	82.6	79.2	20.8
Multi-Tenant TE [32]	22.9	0.96	2.8	93.5	89.0	65.9	14.2
AHP-AI (Proposed)	**20.8**	**0.63**	**2.4**	**95.0**	**93.4**	**58.4**	**11.3**

TABLE III
ROUTING MODEL PERFORMANCE ON USNET (24 NODES, 42 EDGES)-BASELINE

Routing Model	Delay (ms)	PLR (%)	Jitter (ms)	Throughput (Mbps)	ISR (%)	Convergence (ms)	Controller Overhead
IoT E2E [34]	27.8	1.07	3.1	91.2	86.5	72.5	13.2
Static AHP [38]	30.5	1.92	4.2	90.5	77.4	102.3	11.6
Fuzzy AHP [25]	28.9	1.35	4.0	91.0	81.3	93.5	13.5
Deep RL (DQN) [26]	26.3	1.12	3.5	**94.6**	84.1	78.4	18.7
Flow-aware DRL [27]	25.1	0.95	3.2	93.9	85.7	71.5	20.4
Reusable RL [28]	25.9	1.05	3.4	92.8	83.6	76.3	16.9
MARL (Sun et al.) [29]	24.5	0.98	3.1	93.2	86.2	69.7	17.8
MARL (Kang et al.) [30]	24.2	0.93	2.9	93.5	88.0	66.1	17.2
MILP QoS-Slicing [31]	26.1	1.20	3.7	92.1	80.5	85.3	22.6
Multi-Tenant TE [32]	25.7	1.08	3.3	93.0	87.4	70.2	15.9
AHP-AI (Proposed)	**23.4**	**0.72**	**2.8**	93.7	**92.3**	**63.1**	**12.4**

TABLE IV
ROUTING MODEL PERFORMANCE ON OS3E TOPOLOGY (34 NODES, 41 EDGES)

Routing Model	Delay (ms)	PLR (%)	Jitter (ms)	Throughput (Mbps)	ISR (%)	Convergence (ms)	Controller Overhead
IoT E2E [34]	30.6	1.28	3.6	89.1	84.0	77.6	14.7
Static AHP [38]	33.4	2.10	4.5	88.2	74.5	109.6	12.9
Fuzzy AHP [25]	31.3	1.52	4.2	89.6	78.2	100.4	14.9
Deep RL (DQN) [26]	28.5	1.28	3.9	91.4	81.3	85.6	21.4
Flow-aware DRL [27]	27.1	1.09	3.5	90.5	82.7	78.9	23.6
Reusable RL [28]	28.0	1.15	3.7	89.6	80.4	83.5	19.4
MARL (Sun et al.) [29]	26.4	1.05	3.3	90.8	83.9	76.7	20.2
MARL (Kang et al.) [30]	25.7	0.98	3.0	91.2	85.4	72.8	19.4
MILP QoS-Slicing [31]	28.0	1.38	4.1	89.0	76.4	91.3	25.9
Multi-Tenant TE [32]	27.3	1.22	3.5	90.0	83.2	77.9	18.1
AHP-AI (Proposed)	**24.9**	**0.88**	**2.9**	**91.8**	**90.5**	**68.7**	**13.8**

these topologies are presented in Tables IV, and V. As expected, delay, jitter, packet loss, and controller overhead increase with network scale, while throughput and ISR slightly decline due to congestion and routing overhead. However, the proposed AHP-AI model consistently maintains the lowest delay, PLR, jitter, and CPU load across all topologies, validating its scalability and real-time reconfiguration capabilities.

Figure 2 shows the comparison with baseline methods in visual form. The first subplot illustrates latency-related metrics, including E2E delay, jitter, and packet loss ratio (PLR), where lower values indicate better performance. The second subplot focuses on throughput and intent satisfaction ratio (ISR). i.e. how intelligently it satisfies the intent of users. Hence it is mentioned as intelligent selection rate in the graph (e). The third subplot represents control plane overhead via convergence time and CPU usage at the controller, where lower values denote more efficient operation. The proposed AHP-AI model consistently outperforms others across most metrics, demonstrating its effectiveness in balancing QoS and network adaptability.

Figure 2 illustrates the reduction in E2E delay, where the proposed framework achieves a 36% improvement over IoT E2E and around 11% over leading DRL models. This benefit is particularly impactful for real-time applications such as interactive video conferencing and teleoperation, where latency is critical.

Moreover, in terms of packet loss, our system consistently maintains rates below 1%, outperforming both fuzzy-AHP and heuristic-based satellite routing [24]. This improvement stems from the AI modules ability to reroute traffic based on live congestion metrics, avoiding bottlenecks before degradation occurs. Furthermore, jitter control is another area where the proposed method demonstrates strength. With jitter levels reduced to 2.8 ms, it provides superior support for multimedia traffic, outperforming stochastic RL models [28] that lack real-time prioritization. While DRL models like [27] yield marginally better throughput via aggressive exploration, they incur higher convergence time and controller CPU usage. Our

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TABLE V
ROUTING MODEL PERFORMANCE ON DFN TOPOLOGY (58 NODES, 87 EDGES)

Routing Model	Delay (ms)	PLR (%)	Jitter (ms)	Throughput (Mbps)	ISR (%)	Convergence (ms)	Controller Overhead
IoT E2E [34]	33.2	1.42	3.9	87.0	82.3	81.5	15.4
Static AHP [38]	36.8	2.28	4.7	86.1	71.2	115.2	13.8
Fuzzy AHP [25]	34.4	1.68	4.4	87.3	75.6	106.7	15.8
Deep RL (DQN) [26]	30.9	1.36	4.1	90.0	79.2	90.1	22.5
Flow-aware DRL [27]	29.4	1.15	3.7	89.2	80.6	82.4	24.8
Reusable RL [28]	30.2	1.23	3.9	88.1	78.0	87.2	20.3
MARL (Sun et al.) [29]	28.7	1.12	3.5	89.6	81.3	79.1	21.0
MARL (Kang et al.) [30]	27.9	1.06	3.2	90.1	83.2	75.4	20.1
MILP QoS-Slicing [31]	30.0	1.50	4.3	87.1	73.2	97.4	27.2
Multi-Tenant TE [32]	29.2	1.35	3.8	88.3	80.5	82.3	19.2
AHP-AI (Proposed)	**26.4**	**0.95**	**3.0**	**90.7**	**89.2**	**70.6**	**14.5**

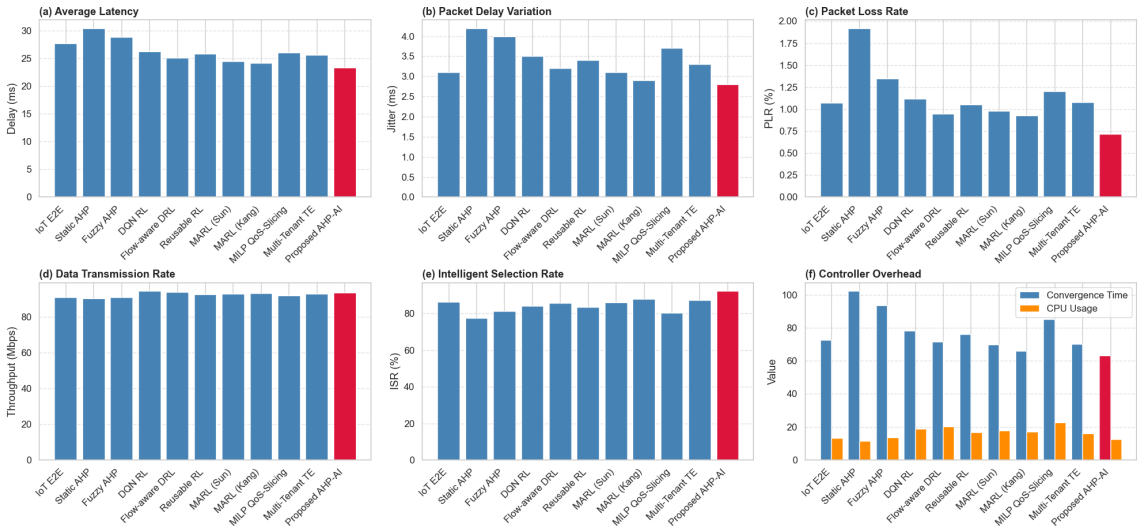

Fig. 2. The comparison includes IoT E2E, Static AHP, RL-Based, Fuzzy-AHP, and the proposed AHP-AI hybrid model.

approach, in contrast, achieves 93.7 Mbps throughput, while converging in 63.1 msideal for multi-tenant and fast-changing environments such as those described in [32]. The ISR metric is particularly significant for IBN scenarios. With a score of 92.3%, the proposed framework surpasses all static and AI-based baselines, showcasing its alignment with user-defined service goals.

This confirms that combining AHP-based prioritization with AI-driven weight adaptation delivers both intent compliance and performance optimization. Furthermore, the controller overhead remains contained at 12.4%, much lower than DRL-heavy models [26], and far more computationally efficient than MILP-based systems [31]. While multi-agent reinforcement learning approaches [29], [30] demonstrate solid adaptability, they often suffer from synchronization overhead and long training timeschallenges that the proposed AHP-AI framework avoids through lightweight inference and transparent decision logic. As evident from Table II to Table V, the proposed model sustains superior trade-offs across performance dimensions, delivering predictable, interpretable, and adaptive behavior in SDN environments. Figure 3 further emphasizes this advantage with visual evidence of consistently favorable metrics across routing scenarios. The heatmap (left) represents a ”best-wins” encoding, where each cell marked with a 1 indicates the best-performing model for a specific metric (i.e., lowest delay, jitter, PLR, convergence time, or CPU usage; highest throughput and ISR). The bar plot (right) summarizes the total number of metric wins achieved by each model. This visualization highlights the superior consistency of the proposed AHP-AI framework, which outperforms or matches state-of-the-art techniques across the majority of performance parameters.

Furthermore, Table II shows the results for Abilene topology, Table III for USNet, Table IV for OS3E, and Table V for DFN. The number of nodes increases from Table III to V. As the complexity of the topology increases we can see a degradation in the performance with respect to delay, jitter, PLR and throughput. This is evident from Table V, where the complexity is at highest and the performance metrics show high delay, jitter, PLR and low throughput as compared to Abilene, USNet and OS3E. However, the suggested method employing AHP and AI shows significantly good performance in terms of the performance metrics.

VI. CONCLUSION AND FUTURE DIRECTIONS

In this paper, we introduced a hybrid intent-based routing framework leveraging SDN that integrates the AHP with

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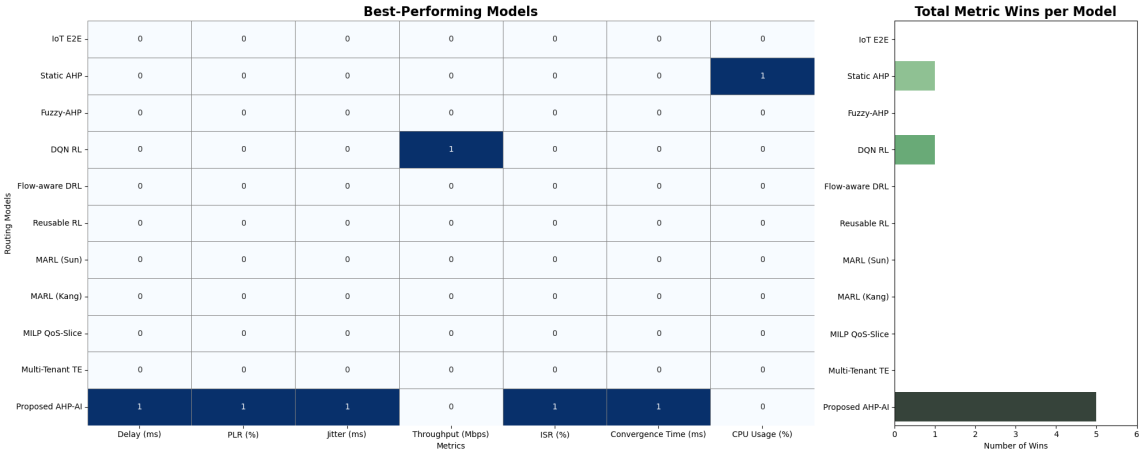

Fig. 3. Binarized performance comparison of baseline routing models across seven key QoS metrics.

lightweight AI optimization to enable adaptive, QoS-aware path selection utilizing IBN. By translating high level service intents leveraging IBN into weighted QoS metrics via AHP and continuously refining those weights through AI-driven feedback, the system achieves a dynamic balance between user policy compliance and real-time network responsiveness. Comprehensive evaluations across diverse topologies including Abilene, USNet, DFN, and OS3E demonstrated that the proposed method consistently outperforms both static and learning-based baselines across key metrics such as delay, jitter, packet loss, throughput, controller efficiency, and ISR. The results affirm that the integration of explainable decision structures with adaptive intelligence not only enhances routing effectiveness but also offers a scalable, interpretable, and a robust foundation using IBN in dynamic SDN environments for future research. The future works include more sophisticated hybrid techniques with the fusion of AI, IBN, SDN and multi-objective techniques for the QoS improvement and optimization

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